

Real Time Full Body Tracking – Full Body, Face & Hand Tracking for Unity

Introduction

Real Time Full Body Tracking is a complete, all-in-one real-time human tracking solution for Unity that combines **Full Body Pose Tracking**, **Hand Tracking**, and **Face Mesh Tracking with animation** into a single, unified plugin.

Built on high-performance RGB-based computer vision, Real Time Full Body Tracking works with any standard camera — **no depth sensors or special hardware required**. It can independently detect and track **face, hands, and full body**, allowing developers to enable or disable each module based on project needs and performance requirements.

Real Time Full Body Tracking is designed for creators building **VTuber systems, virtual avatars, AR/VR experiences, motion-based games, virtual try-on apps, and real-time digital humans**. With plug-and-play prefabs, demo scenes, and inspector-based configuration, you can get realistic avatar tracking running in minutes.

The system supports real-time animation of humanoid avatars (including facial blendshapes), runs smoothly up to **60 FPS**, and works seamlessly with **Ready Player Me avatars**.

Key Features

Core Tracking Features

- **Full Body Pose Tracking** using RGB camera
- **Hand Tracking with 20+ 3D landmarks per hand**
- **Face Mesh Tracking & Facial Animation** (blendshape-based)
- **Independent Module Control** – enable or disable Face, Pose, or Hand tracking
- **Real-Time Tracking** with low latency
- **No Depth Camera Required** – works with standard webcams and mobile cameras

Avatar Animation

- Animate **any humanoid avatar**
- Works out-of-the-box with **Ready Player Me avatars**
- Full body motion replication
- **Up to 180° body rotation tracking**
- Automatic **IK & FK handling** (no manual setup)

- Facial animation via **blendshapes (ARKit compatible)**
- Hand mesh and finger joint animation

Performance & Control

- Adjustable **motion smoothness**
- Configurable **visibility thresholds**
- Auto-scaling avatar based on camera distance
- Multiple model types (Lite / Heavy)
- Supports Camera, Video, and Image input

Platform Support

-  Windows, macOS, Linux (out of the box)
-  Android & iOS (Lite model recommended)

Demo & Samples

- Pose Tracking Demo
- Face Tracking Demo
- Hand Tracking Demo
- Real Time Full Body Tracking Full Tracking Demo
- VTuber Demo Scene

Use Cases

1. **VTuber & Live Streaming Avatars**
 2. **Digital Humans & Virtual Characters**
 3. **Motion-Based Games**
 4. **Gesture-Based UI & Interaction**
 5. **Avatar Motion Capture (Mocap)**
 6. **Education & Training Simulations**
 7. **Virtual Try-On Applications**
 8. **Remote Collaboration with Avatars**
 9. **Fitness & Training Avatars**
 10. **Research & Human-Computer Interaction (HCI)**
 11. **AR/VR Virtual Presence Systems**
 12. **Avatar Puppeteering & Animation Preview**
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Project Setup

Step-by-Step Guide

1. Create a New Scene

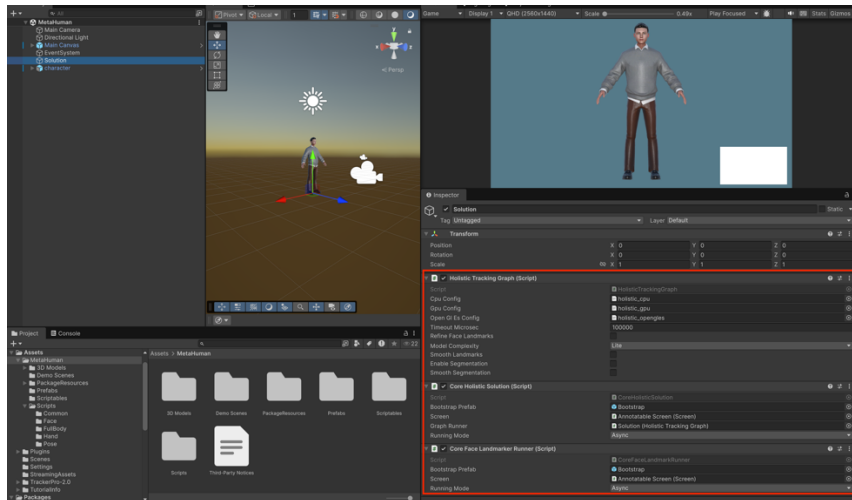
- Open Unity
 - Create a new scene
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2. Add a Camera

- Add either an **Orthographic** or **Perspective** camera
 - Position it to face the avatar
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3. Attach Solution Scripts

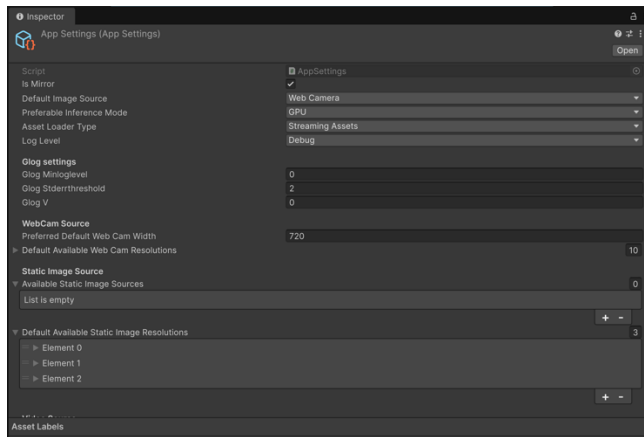
1. Create an empty **GameObject**
2. Attach the following scripts:
 - **Solution**
 - **Graph**
 - **Runner**



4. Configure App Settings

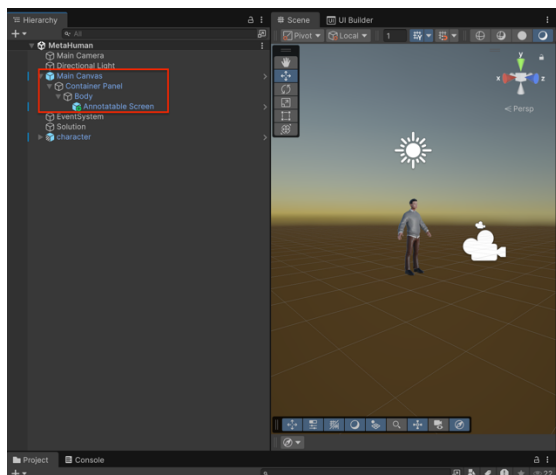
1. Select the **Bootstrap Prefab**

2. Open the **App Settings Scriptable Object**
3. Configure:
 - Input Source (Camera / Video / Image)
 - Model Type (Lite / Heavy)
 - Running Mode
 - Resolution
 - Enable / Disable Face, Pose, Hand Tracking



5. Add Main Canvas

- Drag the **MainCanvas Prefab** into the Scene Hierarchy



Refer to the demo scenes for recommended configurations

Import & Setup Avatar

Avatar Requirements

- Must be **Humanoid Rig**
- Facial animation requires **blendshapes** (ARKit recommended)

Recommended: Ready Player Me Avatar

Download an avatar with facial blendshapes using the link below:

https://models.readyplayer.me/avatar_id.glb?morphTargets=ARKit&textureAtlas=1024

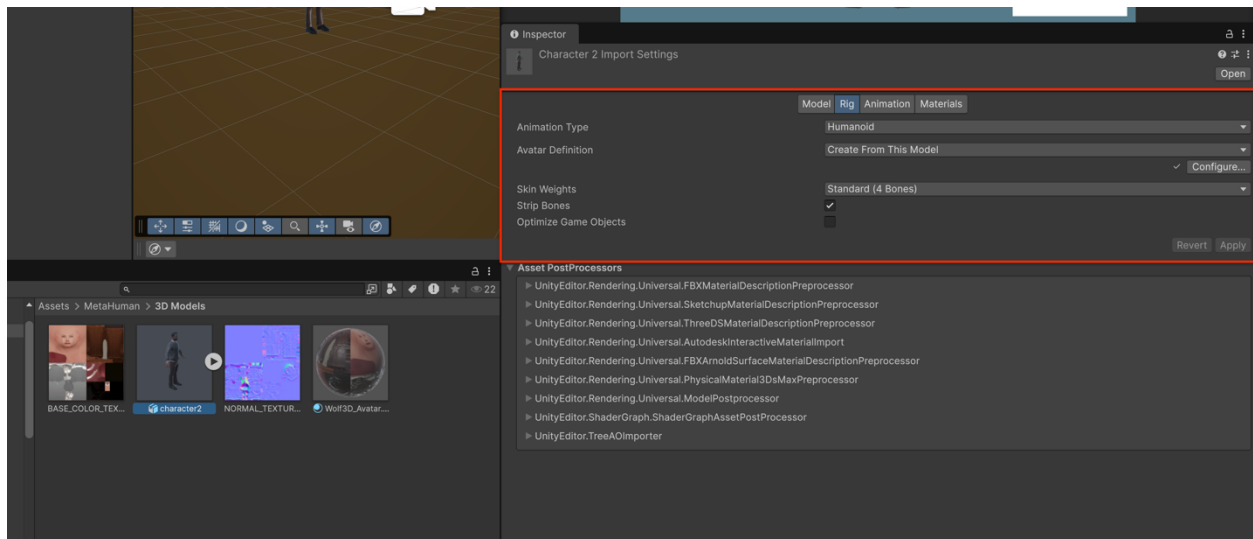
- Convert **GLB** → **FBX** before importing into Unity

You can also download the sample avatar provided by the plugin:

https://drive.google.com/file/d/1KxctzTEvZ7js_Y3c3wn5v1vXsHxmbO4i/view?usp=sharing

Import Avatar into Unity

1. Import the avatar model
2. Select the model → **Rig Tab**
3. Set **Animation Type = Humanoid**
4. Apply changes



Add Avatar to Scene

- Drag the avatar into the scene
- Position it in front of the camera

Attach Tracking Scripts

Face Tracking

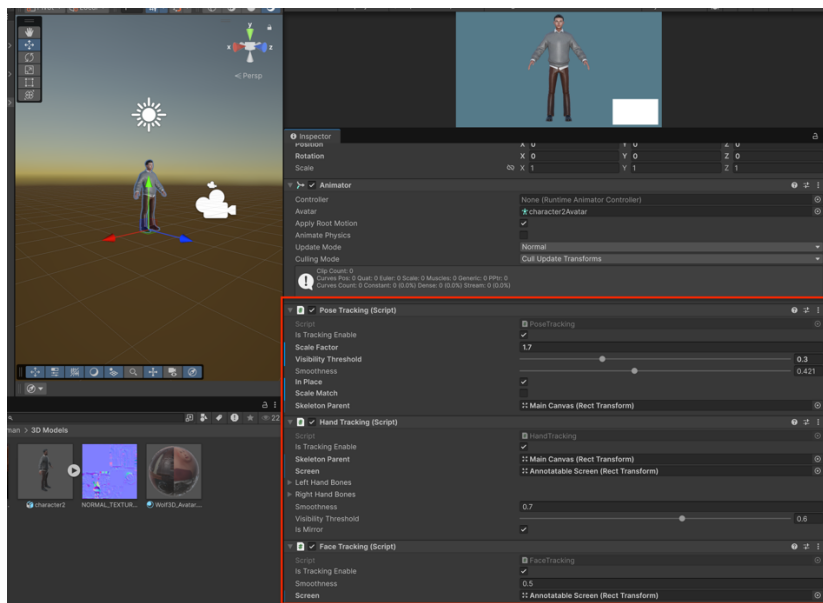
- Add **FaceTracking** script to the avatar
- Assign blendshapes and face settings

Pose Tracking

- Add **PoseTracking** script to the avatar
- Configure body tracking options

Hand Tracking

- Add **HandTracking** scripts for left and right hands
- Assign hand meshes if needed



Common Configuration Fields

Field	Description
Scale Factor	Adjusts avatar scale to match human
Visibility Threshold	Landmark confidence threshold
Smoothness	Controls animation interpolation
In Place	Keep avatar position fixed
Match Scale	Auto-scale based on camera distance
Skeleton Parent	Parent transform for landmarks

Performance Tips

- Use **Low Resolution Input** for better FPS
- Ensure **bright, even lighting**
- Use **Lite models** on mobile devices
- Tune **Visibility Threshold & Smoothness**
- Disable unused tracking modules (Face / Hand / Pose)

Support & Contact

 **Email:** realmoctopus@gmail.com

 **Unity Asset Store Support:** Contact via the Real Time Full Body Tracking Asset Page

If you encounter any issues or need help with setup or customization, feel free to reach out.